

Roberto García Antonio

Contact information

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Education

2023–2025 M.Sc., Mathematics, Universidad Veracruzana

Thesis Title: Reproductive Kernel Hilbert Spaces and Spherical Representations of Groups $SO(n)$ y $U(n)$

Supervisors: Dr. Josué Ramírez Ortega, Dra. Yessica Hernández Eliseo

Abstract: Proved the irreducibility of a representation on a RKHS using a criterion based on the existence of a fixed vector under the action of a closed group.

2019–2023 B.Sc., Mathematics, Universidad Veracruzana

Thesis Title: Álgebras de dimensión finita (finite-dimensional algebras)

Supervisors: Dr. Josué Ramírez Ortega, Dra. Yessica Hernández Eliseo

Abstract: Provided an explicit proof of the Jordan decomposition theorem to characterize the algebra generated by an operator on a finite-dimensional vector space.

Appointments

2023–2025 Teaching Assistant, Universidad Veracruzana

2023–2023 Undergraduate teacher, Universidad Veracruzana

2023–2023 Research Assistant, Dr. Josué Ramírez Ortega SNI 2, Area- Operator algebra

2024-present Analysis seminar, organizer Universidad Veracruzana

Teaching

Teaching Assistant

Conferences

2023	Topology, Universidad Veracruzana, Winter semester	
2024	Differential calculus on several variables	Universidad Veracruzana
2024	Integral calculus on several variables	Universidad Veracruzana
2024	Linear algebra	Universidad Veracruzana
2025	Group theory	Universidad Veracruzana

[1] **Third Colloquium on Lie Theory and Applications, Faculty of Mathematics, Universidad Veracruzana, Mexico- February 2026**

Area: Harmonic analysis

Mini course: [Symmetry and analysis: From Lie groups to spherical harmonics](#)

[2] **58th Congress of the Mexican Mathematical Society, Faculty of Sciences, BUAP, Puebla-October 2026**

Area: Analysis

Talk: [When linear algebra becomes analysis: a look at spectral theorems](#)

[3] **Seminario de Análisis Matemático y Aplicaciones, Universidad Veracruzana, Facultad de Matemáticas, México**

Area: Analysis

talk: Pensando en integración-Integral de Pettis (Thinking About Integration – The Pettis Integral)

[4] **Segundo coloquio de teoría de Lie y aplicaciones, Facultad de Matemáticas, Universidad Veracruzana, México**

Area: Lie Theory

talk: Grupos de Lie y representaciones irreducibles: el grupo $SO(n)$ (Lie groups and irreducible representations: $SO(n)$ group).

[5] **Congreso de la sociedad mexicana de matemáticas, Facultad de Ciencias Exactas, Universidad Juárez del Estado de Durango (UJED), México**

Area: Analysis

talk: Representaciones irreducibles del grupo $SO(n)$ (irreducible representation of the $SO(n)$ group).

[6] **Primer seminario de análisis matemático y aplicaciones UV, Facultad de Matemáticas, Universidad Veracruzana, México**

Area: Analysis

talk: Grupos topológicos con vistas a teoría de representaciones (Topological Groups from the Perspective of Representation Theory).

- [7] **Primer coloquio de teoría de Lie y aplicaciones, Facultad de Matemáticas, Universidad Veracruzana, México**

Area: Lie Theory

talk: Formas bilineales y cuadráticas (Bilinear and quadratics forms).

- [8] **Escuela de verano 2023 CIMAT, CIMAT-Unidad Merida, Mexico**

Area: Operator Theory

Talk: Algebras y una prueba del Teorema de descomposición de Jordan (Algebras and a Proof of the Jordan Decomposition Theorem).

Schools and workshops

- [1] **School of differential geometry and topology**, CIMAT, Guanajuato, 2025.
- [2] **Third school of geometric group theory**, UNAM-CCM, Morelia, 2024.
- [3] **Cimat summer school**, CIMAT, Merida, 2023.
- [4] **Eighth Mexican Congress on Fixed Points and Applications**, CIMAT, Merida, 2023.
- [5] **International Workshop on Operator Theory on Function Spaces**, UNIVERSIDAD VERACRUZANA-CINVESTAV, Veracruz, 2022.

Professional Activities

Science communication

- **Workshop for mathematics education-Colegio las Américas, Xalapa, 2023**

Workshop

Codificando y decodificando imágenes

- **XVI Encuentro xalapeño de física, 2019**

Workshop

¿Qué onda con las ondas?, Grupos LUX

Languages

- Spanish-Native
- English-Advanced, B2 certificated by **TOEFEL IBT**
- German-Beginner

References

Available on request.